LEADING INDICATORS OF CONSUMER CREDIT STRESS: A MACRO ANALYSIS

DS2500 - Team Project Final Report

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1. Introduction

Goal of the Project

To identify which macroeconomic indicators best predict recessions, expressed in terms of constrained credit environments as measured by delinquency rates.

Problem Statement & Background

Credit card delinquencies signal household financial stress, but by the time they rise it may be too late for policymakers to act. Times of sharp credit constraint come with significant hardship for everyday Americans, being unable to pay off lines of credit used to cover everyday expenses or emergency spending. The aggregate of these late payments is known as the delinquency rate. These periods of economic hardship can be tracked and measured by the delinquency rates, but that only captures the effect as it happens, providing policymakers and central bankers with no window to enact meaningful preventative action. If earlier signals of these downturns were identified, and the delayed relationship they have with credit shared, policymakers are given time to mount a defense, reducing the damage recessions have on the economy. Being two economics students passionate about macroeconomics, this felt like a natural place to apply the techniques developed in DS2500 to look for a solution.

The central question of our work is: During the 2008 crisis, the 2020 COVID shock, and the 2021–2023 inflation period, which indicators moved first and had the strongest relationship with delinquencies? We believe the implications of our findings are of interest not only for economics and data science students, but all who want to better understand the dynamics of our modern economy.

Why This Matters

As seen in the most recent economic downturn during the COVID-19 pandemic, recessions affect every aspect of everyday life. Downturns affect employment opportunities, suppress retirement savings

and purchasing power, and greatly contribute to the ongoing affordability crisis. Credit conditions are a very strong contributor to recessions[1], and understanding their movements as it relates to other economic indicators better equips lenders and policymakers to anticipate credit strain before it shows up in the data.

Properly bracing the economy for downturns has significant effects on the lives of millions. During the Great Recession, $414 billion USD was wiped from US GDP with a total of 8.7 million jobs lost[2]. These losses have significant effects on the lives of members of these households, creating periods of despair among families and their communities. Many Americans became incapable of meeting their debt obligations and subsequently lost their homes. An estimated 3.8 million foreclosures occurred between 2007 and 2010. This is a direct result of policy being ill-equipped to respond in a time of dire economic stress. Better understanding the early warning signs of economic strain allows for preventative measures that soften the landing before damage is done.

2. Dataset

Dataset Overview

Source: Federal Reserve Economic Data (FRED) provides over 800,000+ economic time series. Global standard of macro data aggregation. Data series are sourced from 119 different public and private sources, covering areas such as interest rates, employment, and GDP.

1. PCE Health: Personal Consumption Expenditures: Healthcare link
2. PCE Food: Personal Consumption Expenditures: Food link
3. PCE Energy: Personal Consumption Expenditures: Energy link
4. PCE Core: Personal Consumption Expenditures Excluding Food and Energy link
5. Credit Card DR: Delinquency Rate on Credit Card Loans link
6. Consumer Loan DR: Delinquency Rate on Consumer Loans link
7. Mortgage DR: Delinquency Rate on Single-Family Residential Mortgages link
8. VIX: CBOE Volatility Index link
9. Housing Permits: New Privately-Owned Housing Units Authorized link
10. U6 Unemployment Rate: Total Unemployed Plus All Marginally Attached Workers link
11. USREC: NBER-Based Recession Indicators for the United States link

Time Period: 1997 – Present

[1] Kohoe et al. 2020
[2] See Federal Reserve History "*The Great Recession of 2007–09*"

Size: All Quarterly; 104 quarters

Format: CSV format

Data Collection Methodology

Most of our data comes in the form of index tracking price changes for bundles of goods, our PCE metrics, published by the Bureau of Economic Analysis. The Volatility index is published by the Chicago Board Options Exchange, calculated as the square root of the weighted sum of S&P500 Options prices, giving the volatility percentage. PERMITS is the total count of new housing development permits issued, given as a raw monthly count. We use the U-6 measurement here because it better reflects the long-lasting employment hurdles economic downturns create, originally sourced by the Current Population Survey, which uses random sampling to survey approximately 60,000 households monthly. Delinquency rates are reported to the FED by commercial bank call reports submitted quarterly, and the National Bureau of Economic Research (NBER) publishes the official determination of past recessions, displayed in the dataset as a binary variable.

Key Variables

Variable Name	Description	Type	Example Values
pce_health_yoy_2y	Healthcare PCE inflation, lagged 2 years	Numeric	2.1%, 4.3%
vix_6m	CBOE Volatility Index, lagged 6 months	Numeric	15.2, 42.7
permit_yoy_1y	Housing permits YoY % change, lagged 1 year	Numeric	-12.4%, 8.1%
ccard_dr	Credit Card Delinquency Rate (target)	Numeric	1.53%–6.77%
rec_ind	NBER Recession Indicator	Binary	0, 1

Ethical Considerations

Because we are working with national and regional averages, there are no individual privacy concerns with our dataset. A potential bias of this approach of this is the smoothing out of experiences on the outer ends of the sample, specifically the lower end of income distributions where economic strain is felt the strongest. This is even more true for those at the very bottom of the income distribution, who do

not open bank accounts and who are unable to obtain credit lines, forcing our analysis to underrepresent the experiences of those at the most risk. Any conclusions drawn from our analysis risk understating the severity of debt stress on the most economically vulnerable populations and should be kept in mind when discussing policy implications.

3. Methods

Data Preprocessing/Preparation

We chose to perform all transformations internally, converting raw index levels to year-over-year percent changes to capture momentum rather than absolute levels. This preserves replicability and the intuitive understanding of a 1-unit change. Monthly series were resampled to a common quarterly frequency using the last observation of each quarter, ensuring temporal alignment across all indicators before analysis. Mean values were used to form quarterly metrics, besides the binary recession variable which used max to ensure all recessions were properly captured.

Exploratory Data Analysis

Initial exploration of summary statistics and line graphs showed the behavior and bounds of each indicator across 104 quarters spanning three distinct recessions. Transforming raw PCE figures into YoY % change provided better context into relative movements, more suited to understanding how variables move together with lagged effects. Correlation analysis at 6-month, 1-year, and 2-year lags identified healthcare inflation, housing permit activity, and market volatility as the three strongest leading indicators of consumer credit stress. While Mortgage DR holds the highest variance, it is driven up by the 2008 Financial Crisis and not a repeatable trend of the larger US economy, shown in its lower Pearson correlations with our selected features. Credit Card delinquencies have the strongest relationship across variables, respond faster to economic stress than mortgages and consumer loans, and retain higher variance than consumer loans, providing more variation for the models to learn from.

Modeling & Evaluation

We tested predictive power using univariate and multivariate linear regression, then logistic regression to find what features best predict a stressed credit environment, defined as delinquency rates at or above half a standard deviation above the median. Models were validated with K-Fold cross-validation

to account for the small number of quarters available. Without the concern of improper data splitting, OLS allows us to use the full sample to calculate coefficients, R-squared values, and p-values, getting at the heart of our research question

4. Results & Conclusions

KFold cross-validation tests predictive performance across shuffled data splits, while OLS uses the full sample to quantify the statistical relationships between features and target. Together they show both the directional strength of each indicator and how well they explain credit card delinquency rates, with the limitations of each approach discussed further in the limitations section.

Key Findings

Finding 1: Healthcare inflation has the strongest relationship with Credit Card Stress

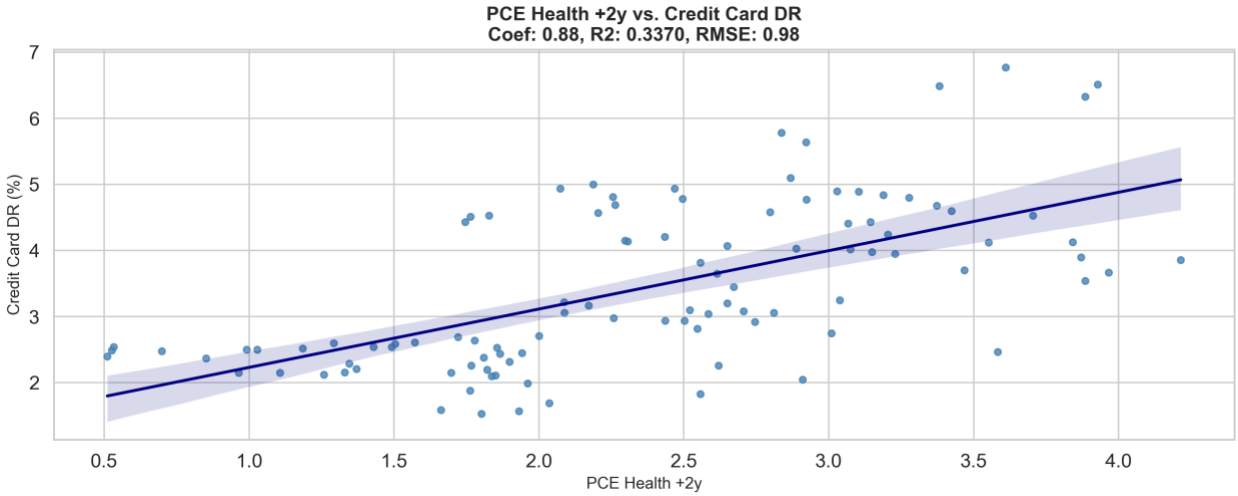

Figure 1: PCE Healthcare YoY% (lagged 2 years) vs. Credit Card Delinquency Rate. Shows a positive linear relationship across the full sample.

A 1% increase in healthcare PCE inflation lagged by 2 years is associated with a 0.88 percentage point increase in credit card delinquency rates, the strongest relationship of all selected features (R-squared = 0.337, Pearson r = 0.62). Healthcare costs take a full 2 years to move through the economy before materializing as missed credit payments.

Finding 2: Housing Permit Activity has a Strong Inverse Relationship

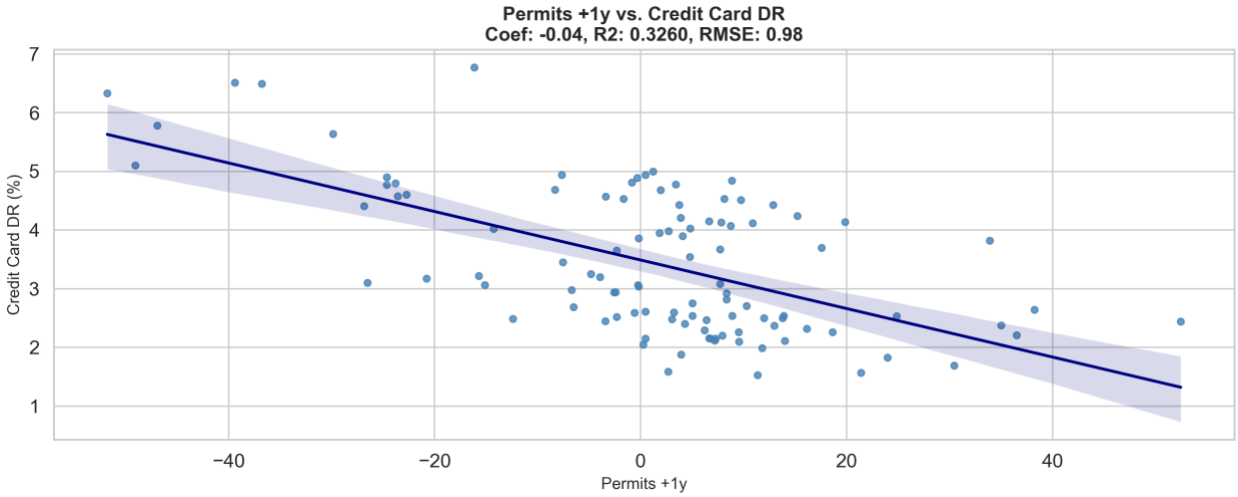

Figure 2: Housing Permits YoY% (lagged 1 year) vs. Credit Card Delinquency Rate. Shows a negative linear relationship, where periods of strong permitting activity precede lower delinquency rates.

A 1% increase in housing permit growth lagged by 1 year is associated with a 0.041 percentage point decrease in credit card delinquency rates (R-squared = 0.326, Pearson r = -0.60), affirming that strong housing activity precedes lower credit strain. This reflects the well-established sensitivity the construction industry has to credit conditions.

Finding 3: The Three Features Jointly Explain a Meaningful Share of Credit Card Delinquency Variation

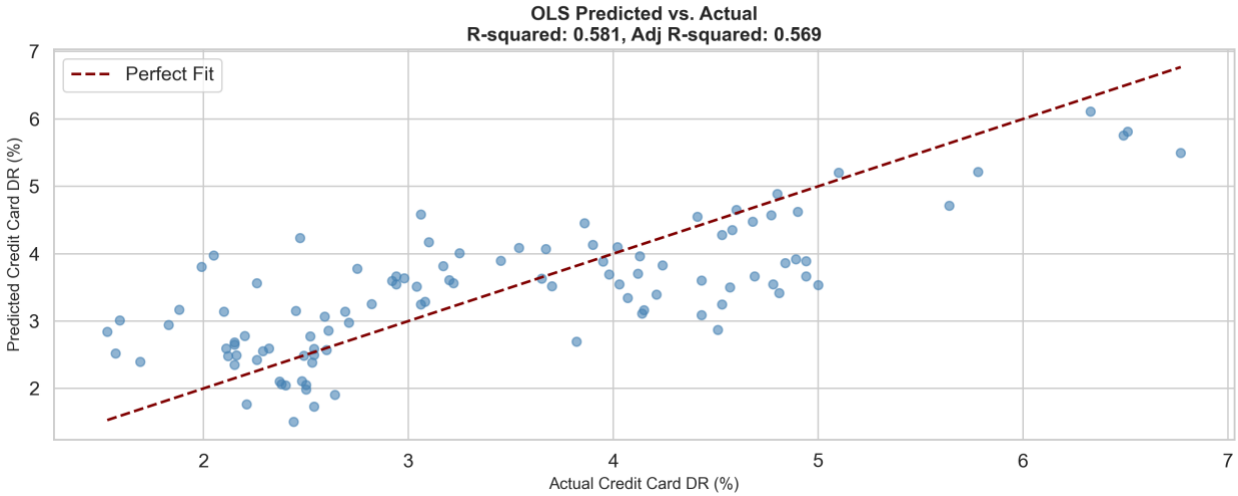

Figure 3: OLS predicted vs. actual credit card delinquency rates across the full sample. Points clustering around the perfect fit line show the model tracks broad movements in credit stress well, with divergence concentrated around the COVID period.

The combined model explains 56.9% of the variation in credit card delinquency rates, high for a 3-variable macroeconomic model. PCE Health ($p < 0.001$), Permits ($p < 0.001$), and VIX ($p = 0.016$) are all statistically significant at the 5% level.

Model Performance

The 3-feature KFold model produced a mean adjusted R-squared of 0.510. Expanding to all 18 features drops it to 0.502, confirming our feature selection captured the strongest signals without overfitting. The logistic regression achieved 72% accuracy against a naive baseline of 60%, showing the three features can meaningfully classify stressed credit environments. All three features remain statistically significant at the 5% level after controlling for recession conditions, meaning they carry independent explanatory power beyond a binary recession proxy.

Metric	Value	Interpretation
Logistic Accuracy	72%	Outperforms naive baseline of 60% (62/104)
Precision (class 1)	68%	When model predicts a stressed environment, correct 68% of the time
Recall (class 1)	60%	Catches 60% of actual stressed periods
F1 Score (class 1)	0.63	Best achieved at median +0.5 SD threshold

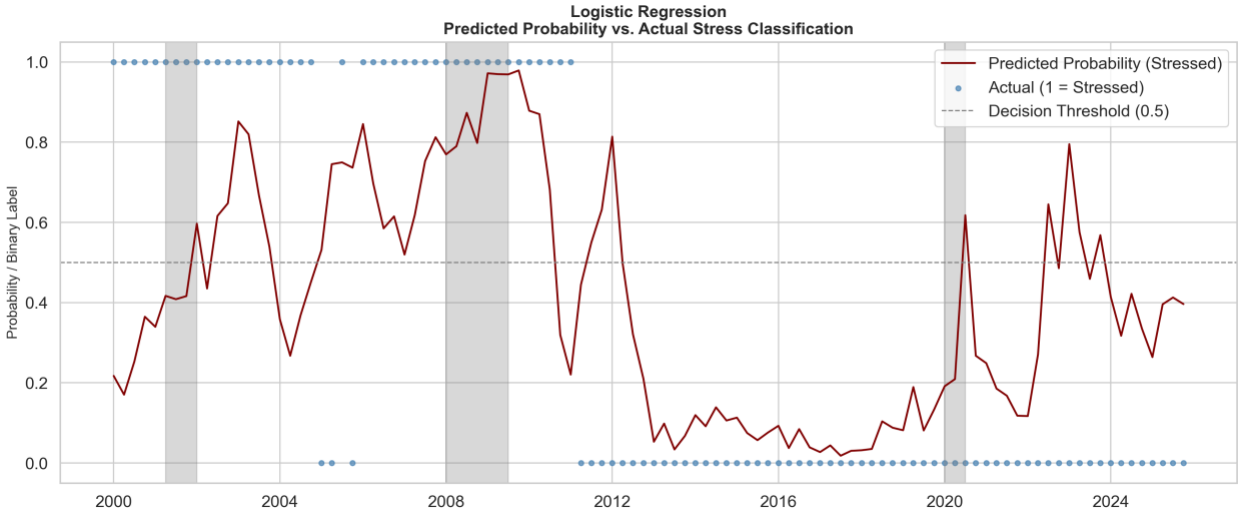

Figure 4: The model correctly identifies stressed credit environments surrounding the 2001 dot-com recession and the 2008 Financial Crisis. Post-COVID, predicted rises toward 0.5-0.8 as macroeconomic indicators deteriorate while credit stress stay suppressed at 0, reflecting a structural break due to government stimulus.

Conclusions

Returning to our original question about which indicators moved first and had the strongest relationship with delinquencies during the 2008 crisis, the 2020 COVID shock, and the 2021-2023 inflation period: healthcare inflation, housing permit activity, and market volatility each show a strong relationship with consumer credit stress, appearing 6 months to 2 years before delinquency rates rise. Among them, PCE Health holds the strongest relationship, followed closely by Permits and a much

smaller effect held by VIX. These relationships hold independent of recession conditions and are specific to consumer revolving credit, sharing similarities with consumer loans but no such relationship with mortgages, where structural dynamics like the MBS collapse fall outside what our indicators capture.

5. Future Work & Limitations

Limitations of Current Analysis

Our datasets span 104 quarters of economic activity and 3 distinct recessions, those being the 2001 dot-com bubble, the 2008 Financial Crisis, and the COVID shock, as well as the unprecedented inflationary period that followed. Percentage and temporal splits cannot fully represent the full range of these economic regimes, nor can K-Fold folds hold all their diversity simultaneously. Massive fiscal intervention during the COVID Pandemic created a structural shift in our observations, suppressing delinquency rates that our models predicted would spike. No pre-COVID trained model can accurately reflect the pandemic to come or the post-shock economy.

The decision to use shuffled K-Folds came when we realized its non-shuffled counterpart would result in similarly isolated training sets. This is not a proper fix, as this approach introduces look-ahead bias, allowing the model to learn from future outcomes within the time series. Because of this, statistics generated from predictive models are not a trustworthy metric, and we opted to include an Ordinary Least Squares approach to answer our research question directly. It is also important to note the model's shortcomings when attempting to capture Mortgage Delinquency rates. Our model captures consumer credit stress but not structural mortgage dynamics such as the MBS collapse surrounding the 2008 recession.

Future Research Directions

Our research has led to clear macroeconomic indicators that can be used by policymakers as warning signs for future credit constraints at varying ranges of time. Future analysis should explore these relationships in the 20th century, as well as extending the pool of features to better represent movements in mortgage delinquencies. Alternative approaches to training predictive models should be explored to determine the effectiveness of recession prediction across downturns with different root causes.

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